

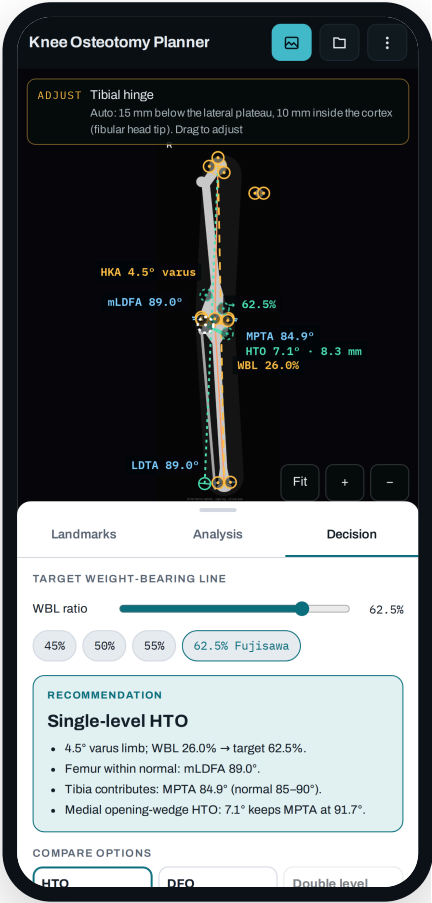
Knee Osteotomy Planner

Alignment analysis on long-leg scannograms, single- vs double-level decision, and wedge sizing for HTO and DFO.

- Place 12 landmarks on a standing long-leg AP film; the app calculates HKA, WBL ratio, MAD, mLDFA, MPTA, JLCA and LDТА.
- Localises the deformity to femur, tibia or joint using Paley's normal ranges.
- Simulates HTO, DFO and double-level plans around real hinge points (Miniaci method) and recommends one, with the reasons.
- Gives the correction angle and the wedge opening height for every cut.
- Subtracts soft-tissue laxity when JLCA is over 3°.
- Works offline. Cases are saved on the phone only.

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|-----------------------|--------------------------------|
| 1 · Before you start | 6 · Case 1: varus, HTO |
| 2 · Calibration | 7 · Case 2: varus + laxity |
| 3 · Placing landmarks | 8 · Case 3: valgus, DFO |
| 4 · Analysis tab | 9 · Case 4: double level |
| 5 · Decision tab | 10 · New case, saving, sharing |

Prepared for Elsayed Elforse, MD · Knee & shoulder arthroscopy
Planning aid only. Not a certified medical device.



The film, the image file, and the screen

Film requirements

- Standing, full weight-bearing, bipedal **long-leg AP** from hip to ankle on one image.
- Patella facing forward. Rotation changes every angle, so repeat the film if the patella is off centre.
- Knee fully extended. A flexion contracture makes the limb look more varus or valgus than it is.
- A **calibration marker** (25 mm metal ball) taped at knee level at the depth of the bone. See section 2.

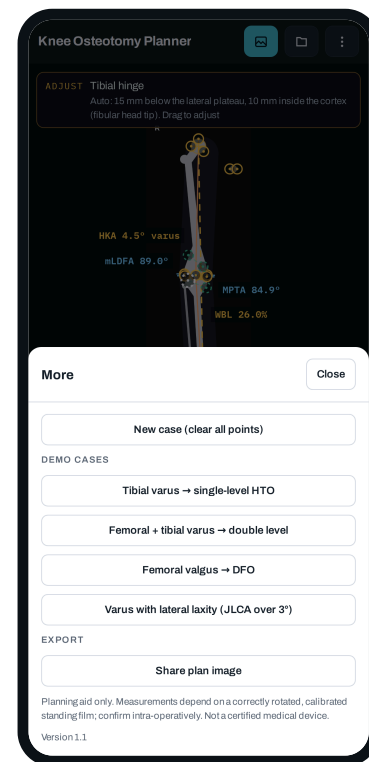
Getting the image into the app

- 1 In PACS, export the long-leg image as **JPEG or PNG** at full resolution. DICOM files cannot be opened directly.
- 2 Send it to your phone (WhatsApp to yourself, email, or cloud drive) and save it to the gallery or Files.
- 3 In the app, tap the **image button** (top right, teal) and choose the file. You can also photograph the PACS screen, but hold the phone square to the monitor and include the ball.

Screen layout

- **Top:** the X-ray viewer. Pinch to zoom, drag empty space to pan, **Fit** resets the view.
- **Bottom panel:** three tabs, **Landmarks**, **Analysis** and **Decision**. Tap the grey handle to make the panel taller.
- **Folder button:** saved cases. **⋮ menu:** new case, demo cases, share plan image.
- The yellow bar at the top of the viewer always says which point to tap next.

Try the four demo cases first (⋮ → Demo cases). They are the case examples in this guide, so you can compare the screen with each page.



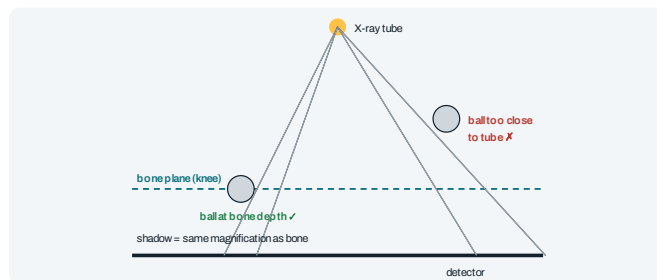
⋮ menu with the demo cases.

How to calibrate

Angles (HKA, mLDFA, MPTA, JLCA) do not depend on calibration. Everything measured in **millimetres** does: the MAD, the hinge-to-entry length and the **wedge opening height**. A 10% calibration error turns a true 10 mm opening into 11 mm, so calibrate every film.

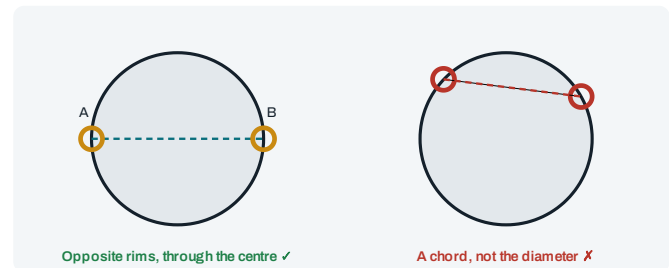
A · Choosing and placing the marker

- **Best:** a metal calibration ball of known diameter, usually 25 mm. A sphere looks the same size whatever its rotation.
- Tape it to the skin **at the level of the knee joint line**, on the medial or lateral side, so it lies at about the same distance from the detector as the bone.
- A radiopaque ruler taped along the limb at bone depth also works. Use two marks a known distance apart, e.g. 100 mm.
- **Do not rely on the PACS scale bar or the DICOM pixel size.** They are measured at the detector, not at the bone, so they ignore magnification and overestimate every millimetre value.



B · Calibrating in the app

- 1 Open the image. The **Landmarks** tab opens with **Marker edge A** selected.
- 2 **Pinch to zoom** on the ball until it fills about a third of the screen.
- 3 Tap the **left rim** of the ball on a horizontal line through its centre. Tap the edge of the metal, not the blurred halo.
- 4 The app moves to **Marker edge B**. Tap the **opposite rim** on the same line.
- 5 Under **Calibration**, set **Marker length** to the true diameter (25 mm by default). For a ruler, enter the distance between your two marks.
- 6 Refine: drag either point. A magnifier appears so your finger does not hide the rim.
- 7 The scale appears next to the field, e.g. 0.595 mm/px. Tap **Fit** and carry on with the hip.



C · Checking it

Drag A and B to the top and bottom rims of the ball and read the mm/px again. The two values should agree within 2%. Put the points back on the horizontal diameter afterwards.

No marker on the film? The app estimates the scale from a 75 mm plateau width and shows a warning. Angles are unaffected; treat mm values as rough.

IF SOMETHING LOOKS WRONG

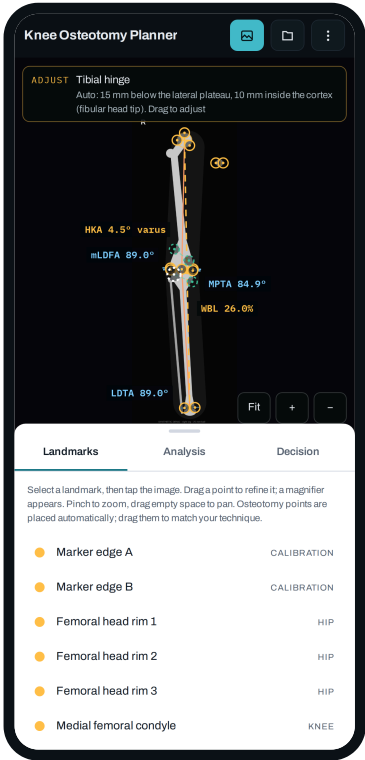
WHAT YOU SEE	LIKELY CAUSE	FIX
Wedge sizes look too big for the angle	Ball taped on the side nearest the X-ray tube, or the PACS scale bar used instead of a ball	Place the ball at bone depth; re-tap the ball edges
mm/px changes when you re-tap	Edges tapped on the blurred halo, or not on the diameter	Zoom in until the ball fills a third of the screen; tap the rim on a line through the centre
Horizontal and vertical diameters disagree by more than 2%	Image stretched on export, or the marker is not a sphere	Re-export at original size; use a ball, not a disc or coin

Twelve points, in order

- 1 Select a landmark in the list (or just follow the yellow prompt), then **tap it on the image**. The app moves to the next one automatically.
- 2 To **refine** a point, drag it. A 3× magnifier appears in the corner.
- 3 To **re-place** a point, select it in the list and tap the new spot.
- 4 Zoom in for the knee and ankle points; accuracy there drives the plan.

LANDMARK	REGION	HOW TO PLACE IT
Marker edge A/ B	Calibration	Opposite rims of the calibration ball across its diameter (see Calibration).
Femoral head rim 1–3	Hip	Three points spread around the femoral head outline. The app fits a circle and uses its centre as the hip centre.
Medial / lateral femoral condyle	Knee	The most distal point of each condyle. Together they define the distal femoral joint line (for mLDFA).
Knee centre	Knee	Midpoint between the tibial spines at the top of the notch. Shared by the femoral and tibial mechanical axes.
Medial / lateral plateau edge	Knee	The rims of the tibial plateau. They define the tibial joint line (MPTA) and the 0–100% scale for WBL.
Talar dome medial / lateral	Ankle	The two shoulders of the talar dome. Their midpoint is the ankle centre; the line between them gives LDFA.
Femoral hinge / cut entry	Automatic	Placed automatically: hinge 28 mm above the medial condyle, 15 mm inside the cortex; entry 65 mm above the lateral condyle. Drag to match your cut.
Tibial hinge / cut entry	Automatic	Placed automatically: hinge 15 mm below the lateral plateau, 10 mm inside the cortex (fibular head tip level); entry 38 mm below the medial plateau. Drag to adjust.

Osteotomy points appear as **dashed green circles**. They follow the anatomy until you drag them; after that they stay where you put them. **Reset osteotomy points to automatic** under the list undoes your changes. The wedge height depends on the hinge-to-entry length, so place the hinge and entry where you will actually cut.



Landmarks tab. Filled yellow = placed, dashed green = automatic.

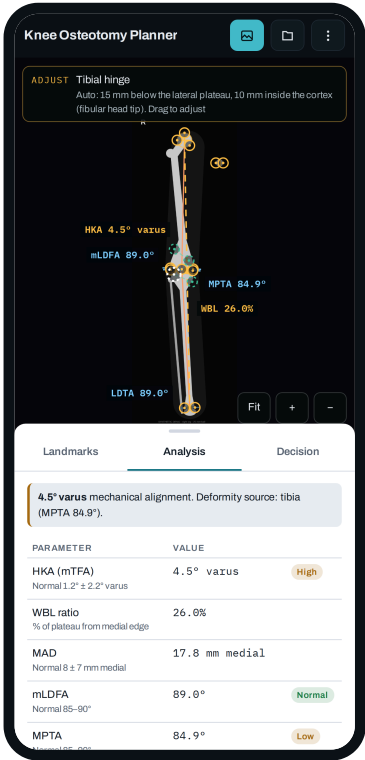
Measurements and where the deformity is

As soon as the hip, knee and ankle points are placed, the Analysis tab shows every parameter with a **Normal** / **Low** / **High** label. The summary line at the top names the deformity source: femur, tibia or joint.

Reading the viewer

- **Yellow dashed line:** mechanical axis (hip centre → ankle centre); the dot on the plateau is the WBL.
- **Orange lines:** femoral and tibial mechanical axes.
- **Blue lines:** joint lines (distal femur, tibial plateau, ankle).
- **Green:** planned axis, planned WBL and wedges (Decision tab only).

Reference values after Paley. JLCA comes from the standing film, so it includes soft-tissue laxity; the Decision tab uses it for the laxity subtraction.



Analysis tab for Case 1.

Parameters

PARAMETER	DEFINITION USED BY THE APP	NORMAL
HKA (mTFA)	Angle between femoral (hip → knee) and tibial (knee → ankle) mechanical axes	1.2° ± 2.2° varus
WBL ratio	Where the hip–ankle line crosses the plateau, from medial edge (0%) to lateral edge (100%)	about 40–60%; osteoarthritis targets 50–62.5%
MAD	Perpendicular distance from knee centre to the hip–ankle line (needs calibration)	8 ± 7 mm medial
mLDFA	Lateral angle between femoral mechanical axis and distal femoral joint line	85–90°
MPTA	Medial angle between tibial mechanical axis and tibial plateau line	85–90°
JLCA	Angle between femoral and tibial joint lines; shows which side is open	0–2°
LDTA	Lateral angle between tibial mechanical axis and ankle joint line	86–92°
Joint line vs limb axis	Tilt of the plateau from a line perpendicular to the hip–ankle axis	Flag above 5°

How the app chooses single or double level

Target

The target WBL defaults to **62.5%** (Fujisawa) for varus knees and **50%** for valgus knees. Change it with the chips (45, 50, 55, 62.5%) or the slider; every plan updates at once.

Decision steps

1. **Direction.** Target above current WBL → valgus-producing correction is needed (varus knee), below → varus-producing (valgus knee).
2. **Localise the deformity.** Femur abnormal if mLDFA is outside 85–90° in the direction of the deformity; tibia abnormal if MPTA is outside 85–90°.
3. **Simulate three plans** by rotating the distal segment around each hinge until the WBL reaches the target: HTO alone, DFO alone, and double level.
4. **Check each plan** against your limits (table below). A breach of an MPTA or mLDFA limit marks the plan “Exceeds limits”.
5. **Recommend:**
 - both bones abnormal → **double level**;
 - one bone abnormal → **single level at that bone** if within limits, otherwise double level;
 - neither clearly abnormal → HTO for varus or DFO for valgus if within limits, otherwise the other single level, otherwise double level.
6. **Double-level split:** femur corrected to mLDFA 87°, tibia takes the rest. If the tibia would still pass the MPTA limit, it is capped there and the femur takes more.

Wedge opening height

For each cut the app measures **L**, the distance from hinge to entry point, and the correction angle **θ**. The opening height is the gap between the two cut surfaces at the entry cortex:

$$\text{opening height} = 2 \cdot L \cdot \sin(\theta/2)$$

The classic table value $L \cdot \tan \theta$ is shown next to it; the two differ by under 0.5 mm below 15°. For closing wedges the same number is the **wedge base** to resect.

Example (Case 1): $L = 67 \text{ mm}$, $\theta = 7.1^\circ \rightarrow 2 \times 67 \times \sin 3.55^\circ = 8.3 \text{ mm}$.

Soft-tissue laxity (JLCA over 3°)

When JLCA is over 3° and opens on the convex side (lateral in varus, medial in valgus), the app subtracts $(\text{JLCA} - 2^\circ) \div 2$ from the bony correction. In a double level it comes off the tibial cut. The plan card shows the angle before and after, e.g. $11.8^\circ - 1.3^\circ$.

Opening or closing

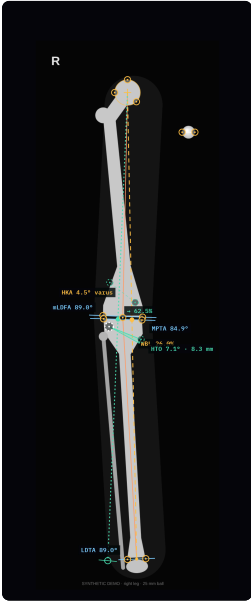
The app names each cut from where you put the hinge. Hinge lateral on the tibia → **medial opening-wedge HTO**; hinge medial on the femur in a varus knee → **lateral closing-wedge DFO**. Move the hinge to the other cortex to plan the other technique.

DECISION LIMITS (EDITABLE UNDER “DECISION LIMITS” AT THE BOTTOM OF THE TAB)

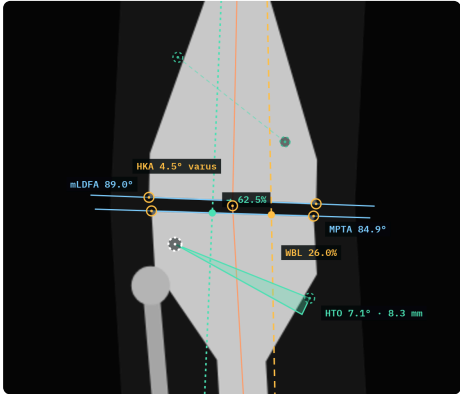
LIMIT	DEFAULT	WHY
Max planned MPTA	95°	Single HTO above this creates excessive joint-line obliquity; widely used limit
Min planned MPTA	85°	For varus-producing HTO in valgus knees
Min / max planned mLDFA	85° / 92°	For DFO; keeps the femoral joint line near normal
Max joint line vs axis	5°	Caution flag only; does not change the decision
Max opening gap	15 mm	Caution flag: hinge fracture risk, graft, patellar height
Double level: femur to	mLDFA 87°	How much the femur takes before the tibia takes the rest
JLCA subtraction trigger	over 3°	Subtract $(\text{JLCA} - 2^\circ) \div 2$ when the open side is the convex side

Varus knee, tibial deformity

Result: Single-level medial opening-wedge HTO. Medial compartment overload with a proximal tibial varus. The femur is normal, so the deformity sits where the osteotomy goes.



Full leg: yellow dashed = pre-op axis, green dashed = planned axis.



Knee close-up: green wedges show each planned cut, labelled with angle and opening height.

PARAMETER	PRE-OP	PLANNED
HKA	4.5° varus	2.3° valgus
WBLratio	26.0%	62.5%
MAD	17.8 mm medial	9.2 mm lateral
mLDFA	89.0°	89.0°
MPTA	84.9°	91.7°
JLCA	0.4° lateral open	0.4°
Jointline vs axis	2.9°	0.6°

PLANNED CUTS

CUT	ANGLE	WEDGE	HINGE → ENTRY
Medial opening-wedge HTO	7.1°	8.3 mm	67 mm

OPTIONS COMPARED (TARGET WBL 62.5%)

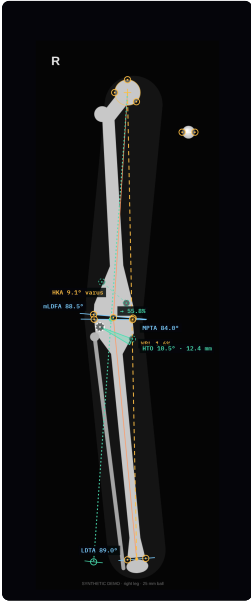
OPTION	ANGLE	WEDGE	STATUS
HTO only	7.1°	8.3 mm	Recommended MPTA 91.7°, mLDFA 89.0°
DFO only	7.4°	8.1 mm	Exceeds Planned mLDFA 82.1° is below 85°
Double level	Not needed: one level carries the correction		

TEACHING POINTS

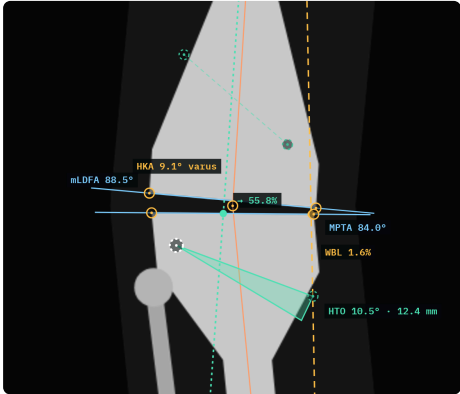
- MPTA 84.9° is the only abnormal angle, so the correction belongs in the tibia.
- A DFO would reach the same target but drives mLDFA to 82.1° and tilts the joint line 6.3°. The app marks it “Exceeds limits”.
- The 7.1° opening gives an 8.3 mm gap at the medial cortex, 67 mm from the hinge. Planned MPTA 91.7° is well under the 95° limit.

Varus knee with lateral laxity (JLCA 4.6°)

Result: Single-level HTO after JLCA subtraction. Same tibial varus pattern, but the standing film shows a lateral joint opening of 4.6°. Part of the malalignment is soft tissue, not bone.



Full leg: yellow dashed = pre-op axis, green dashed = planned axis.



Knee close-up: green wedges show each planned cut, labelled with angle and opening height.

PARAMETER	PRE-OP	PLANNED
HKA	9.1° varus	1.0° valgus
WBLratio	1.6%	55.8%
MAD	36.2 mm medial	4.1 mm lateral
mLDFA	88.5°	88.5°
MPTA	84.0°	94.1°
JLCA	4.6° lateral open	4.6°
Jointline vs axis	1.6°	3.6°

PLANNED CUTS

CUT	ANGLE	WEDGE	HINGE → ENTRY
Medial opening-wedge HTO	10.5°	12.4 mm	67 mm

OPTIONS COMPARED (TARGET WBL 62.5%)

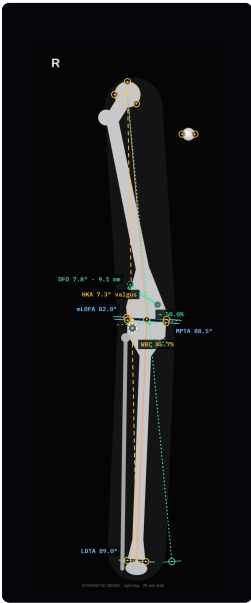
OPTION	ANGLE	WEDGE	STATUS
HTO only	10.5°	12.4 mm	Recommended MPTA 94.1°, mLDFA 88.5°
DFO only	11.0°	12.2 mm	Exceeds Planned mLDFA 78.2° is below 85°
Double level	Not needed: one level carries the correction		

TEACHING POINTS

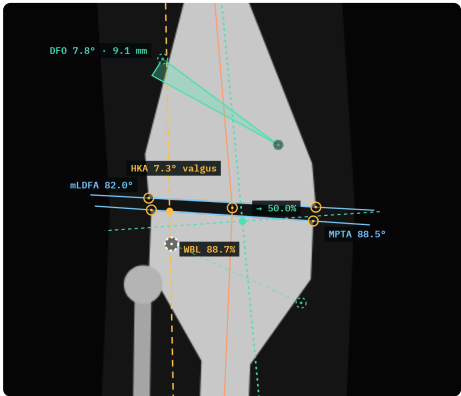
- JLCA is over 3° and opens laterally (the convex side of a varus knee), so the app subtracts $(4.6 - 2) \div 2 = 1.3^\circ$ from the bony correction.
- Bony correction $11.8^\circ - 1.3^\circ = 10.5^\circ$. Opening height falls from about 13.9 mm to 12.4 mm.
- Without the subtraction the planned MPTA would be about 95.4° , over the limit, and the app would recommend a double level. With it, MPTA stays at 94.1° and a single HTO is enough.
- The planned WBL on this film shows 55.8%, not 62.5%. The rest of the shift is expected when the lateral soft-tissue gap closes under load.

Valgus knee, femoral deformity

Result: Single-level lateral opening-wedge DFO. Lateral compartment overload with a valgus distal femur (mLDFA 82°). The tibia is normal.



Full leg: yellow dashed = pre-op axis, green dashed = planned axis.



Knee close-up: green wedges show each planned cut, labelled with angle and opening height.

PARAMETER	PRE-OP	PLANNED
HKA	7.3° valgus	0.0°
WBLratio	88.7%	50.0%
MAD	28.9 mm lateral	0.2 mm medial
mLDFA	82.0°	89.3°
MPTA	88.5°	88.5°
JLCA	0.8° medial open	0.8°
Jointlinevs axis	5.0°	1.5°

PLANNED CUTS

CUT	ANGLE	WEDGE	HINGE → ENTRY
Lateral opening-wedge DFO	7.8°	9.1 mm	66 mm

OPTIONS COMPARED (TARGET WBL 50%)

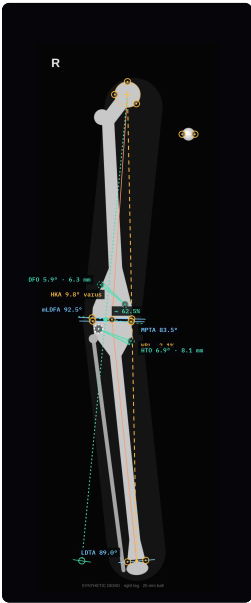
OPTION	ANGLE	WEDGE	STATUS
HTO only	7.5°	8.6 mm	Exceeds Planned MPTA 81.3° is below 85°
DFO only	7.8°	9.1 mm	Recommended MPTA 88.5°, mLDFA 89.3°
Double level	5.3° + 2.4°	6.2 + 2.7 mm	Within limits MPTA 86.2°, mLDFA 87.0°

TEACHING POINTS

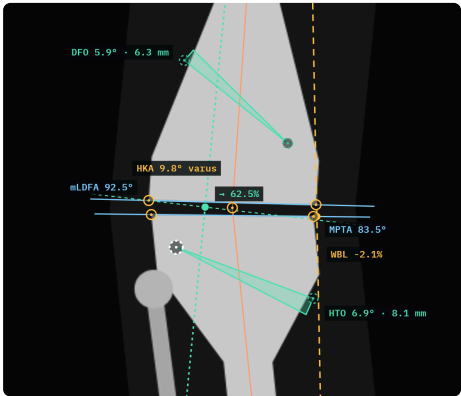
- For valgus knees the app defaults the target to 50% (neutral). Change it with the chips or slider if you prefer slight undercorrection.
- mLDFA 82.0° is the abnormal angle, so a varus-producing DFO is recommended: 7.8° lateral opening wedge, 9.1 mm gap.
- A medial closing-wedge HTO would bring the line to 50% but leave MPTA at 81.3° and a joint line tilted 8.8°. That is the classic mistake of correcting a femoral valgus in the tibia.

Varus knee, femoral and tibial deformity

Result: Double-level osteotomy (lateral closing DFO + medial opening HTO). Severe varus (9.8°) with both mLDFA 92.5° and MPTA 83.5° abnormal. The weight-bearing line passes outside the medial plateau (WBL -2.1%).



Full leg: yellow dashed = pre-op axis, green dashed = planned axis.



Knee close-up: green wedges show each planned cut, labelled with angle and opening height.

PARAMETER	PRE-OP	PLANNED
HKA	9.8° varus	2.4° valgus
WBLratio	-2.1%	62.5%
MAD	39.0 mm medial	9.4 mm lateral
mLDFA	92.5°	87.0°
MPTA	83.5°	90.2°
JLCA	0.8° lateral open	0.8°
Jointlinevs axis	1.8°	1.0°

PLANNED CUTS

CUT	ANGLE	WEDGE	HINGE → ENTRY
Lateral closing-wedge DFO	5.9°	6.3 mm	61 mm
Medial opening-wedge HTO	6.9°	8.1 mm	68 mm

OPTIONS COMPARED (TARGET WBL 62.5%)

OPTION	ANGLE	WEDGE	STATUS
HTO only	12.5°	14.8 mm	Exceeds Planned MPTA 95.6° is above 95°
DFO only	13.1°	14.0 mm	Exceeds Planned mLDFA 80.2° is below 85°
Double level	5.9° + 6.9°	6.3 + 8.1 mm	Recommended MPTA 90.2°, mLDFA 87.0°

TEACHING POINTS

- Both bones are abnormal, so the app recommends double level straight away.
- A single HTO would need 12.5° with a 14.8 mm gap and would push MPTA to 95.6°, over the 95° limit. A single DFO would need 13.1° and would leave mLDFA at 80.2°.
- Double level: femur first to mLDFA 87° (5.9° lateral closing wedge, 6.3 mm base), then the tibia takes the rest (6.9° medial opening, 8.1 mm). Planned MPTA 90.2°, joint line within 1° of perpendicular.

New cases, closing, saving and sharing

Starting a new case

- 1 **Save the current case first** if you want to keep it: folder button → type initials → **Save**. Only the case on screen is kept automatically, and opening a new X-ray replaces it.
- 2 **New patient:** tap the teal **image button** and choose the new X-ray. All points are cleared and the app starts at Marker edge A.
- 3 **Same X-ray, start again:** ⋮ menu → **New case (clear all points)**. The image stays and every point is removed.

Closing the app

- Nothing needs saving before you leave. Press Home or swipe away and the app keeps your work.
- **Android, to close it fully:** open recent apps (swipe up from the bottom and hold, or tap the square button) and swipe Osteotomy Planner away.
- **iPhone:** swipe up from the bottom and pause, then flick the app card up.
- When you open it again, the last case on screen comes back exactly as you left it.

Saving

- The current case is **saved automatically**; closing the app returns you to it.
- To keep a case: tap the **folder button**, type initials and side (e.g. “AS right knee”) and tap **Save**. Open or delete it from the same list; delete asks you to tap twice.
- Cases stay on the phone only. Nothing is uploaded. Use initials, not full names.
- Updating the app by installing a new APK over the old one keeps your cases. Uninstalling deletes them.

Sharing

- **Share image** (Decision tab or ⋮ menu) sends the X-ray with the plan overlay plus a text summary to WhatsApp, email or the gallery.
- **Copy summary** copies the recommendation, cuts and pre/post values as text for your notes or the operation record.

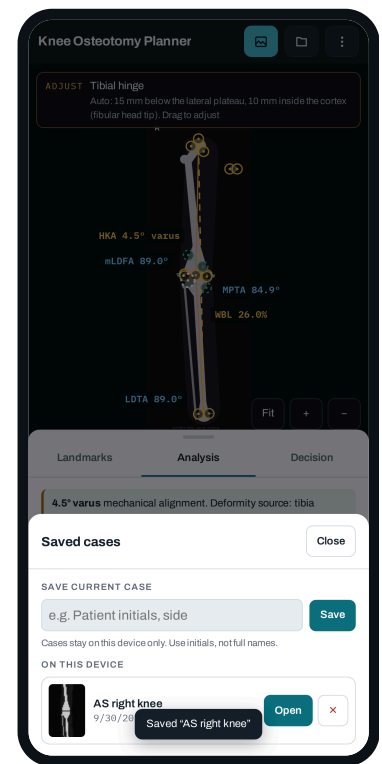
Updates

New versions appear at github.com/eelforse-blip/Knee-planner → Releases → latest. Download **OsteotomyPlanner.apk** and install it over the current app.

Limitations

- The plan is only as good as the film: rotation, flexion and a badly placed marker all change the result.
- The model is 2-D (coronal plane). It does not plan tibial slope, rotation or patellar height.
- Default limits come from widely used thresholds (MPTA 95°) and Paley's normal ranges. Adjust them to your own practice.
- The demo cases use synthetic images, not patient films.

Planning aid only. Not a certified medical device. Confirm the correction on calibrated films and intra-operatively (alignment rod, fluoroscopy).



Saved cases sheet.